

SHAUN DIAS

London, United Kingdom | +44 7733 759527 | md263401@gmail.com

[LinkedIn](#) / [GitHub](#) / [Portfolio](#)

PROFILE

MSc Artificial Intelligence student at King's College London with a First-Class B. Tech in Computer Science & Engineering and hands-on experience in machine learning, computer vision, LLMs, robotics and embedded systems. Experienced in Python, PyTorch, TensorFlow, YOLO, Arduino and ESP32, with internship experience developing AI prototypes, data pipelines and model evaluation workflows. Experienced in building real-world AI and robotics systems, leading technical teams, and communicating technical concepts clearly.

TECHNICAL SKILLS

Programming: Python, C/C++, HTML, SQL

AI / Machine Learning: PyTorch, TensorFlow, OpenAI API, YOLO, Computer Vision, Deep Learning, LLMs

Data & Scientific Computing: Pandas, NumPy, Matplotlib, Data Analysis, Data Preprocessing

Databases: MySQL, Database Management Systems (DBMS), Database Design

Cloud, Systems & DevOps: AWS, Linux, Git/GitHub, GCP

Software & Web Development: Flask, Full-Stack Development, Authentication, HTML

Computer Science Foundations: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Software Engineering

Robotics & Embedded Systems: Arduino, ESP32, Raspberry Pi, Sensor Integration, Servo & Motor Control, Bluetooth Control, Betaflight, PID Tuning

Tools & Methodologies: Jira, Kanban, Excel, Technical Documentation, Debugging

Core Strengths: Problem Solving, Hardware/Software Troubleshooting, Technical Communication, Team Leadership, Cross-Functional Collaboration

PROFESSIONAL EXPERIENCE

Artificial Intelligence Intern — ScholarRankAI (Remote, India) May 2025 – Aug 2025

- Architected AI prototypes and performance dashboards for machine-learning benchmarking initiatives.
- Conducted systematic error analysis and workflow optimization, improving model efficiency.
- Led research into ethical AI usage and best practice.

Tech stack: Python, PyTorch/TensorFlow, dashboarding, benchmarking pipelines

AI/ML Engineering Intern — Ecode Networks Pvt Ltd (India, London, Remote) Apr 2024 – Jul 2024

- Developed and fine-tuned Large Language Models (LLMs) for enterprise applications, supporting a UK-market product launch.
- Designed data pipelines for text processing and model output evaluation.
- Applied optimization strategies that improved accuracy while reducing inference latency.

Tech stack: Python, LLM fine-tuning, data pipeline engineering, model evaluation

Operations & Administration Lead — Stenosis Institute, India May 2024 – Jul 2024

- Managed computer-based administrative systems and workflows in a dynamic operational environment.
- Drafted and edited professional business correspondence.
- Taught English vocabulary and grammar one-to-one — practical experience explaining technical and unfamiliar material clearly.

TECHNICAL PROJECTS

Neural Network Layers from Scratch — Deep Learning

- Implemented fundamental neural-network layers from scratch to develop a deeper understanding of how neural networks operate internally.
- Explored the structure and behavior of individual layers and how data flows through a neural network.
- Used the implementation to strengthen understanding of the foundations behind higher-level deep-learning frameworks.

Technologies: Python, NumPy, Neural Networks, Deep Learning

Vision Guard — Real-Time Micro-Sleep Detection for Drivers

- Built a real-time computer-vision system using facial tracking and YOLO object detection to identify driver fatigue and micro-sleep events.
- Integrated non-invasive camera-based monitoring with adaptive multi-channel alerting (visual, auditory, haptic).

Tech stack: Python, YOLO, real-time facial tracking, embedded alert hardware

Spider Robot — Arduino

- Designed and built a multi-legged robot using Arduino, a motor driver shield, and servo mechanisms.
- Programmed autonomous and Bluetooth-controlled movement; iteratively debugged mechanical and control-logic faults on the physical unit.

Tech stack: Arduino (C/C++), motor driver shield, servos, Bluetooth control

Drone Development and Configuration

- Configured flight-control systems in Betaflight and tuned PID parameters for stability and control accuracy.
- Built and field-tested multiple drone platforms (DJI, Tryo99, Frog frames), diagnosing real-world flight issues under live conditions.

Tech stack: Betaflight, PID tuning, flight-controller firmware, hardware diagnostics.

Wi-Fi Network Analyzer — ESP32-C6

- Developed a compact embedded tool using ESP32-C6 and an OLED display to scan and visualize nearby Wi-Fi networks in real time.

Tech stack: ESP32-C6, embedded C/C++, OLED display integration

IoT-Based Smart Traffic Management System

- Implemented graph algorithms (Prim's, Kruskal's, Dijkstra's) for route optimization and intelligent traffic control.
- Developed a real-time IoT simulation for traffic-flow monitoring and adaptive signal control.

Tech stack: Python, graph algorithms, IoT simulation

Enterprise Banking ERP System

- Built a full-stack ERP system with user/admin dashboards, secure authentication, and payment gateway integration.
- Implemented audit logging and database security layers to standard banking compliance.

Tech stack: Full-stack web development, MySQL, secure authentication, audit logging

EDUCATION

MSc Artificial Intelligence — King's College London

Sept 2026 – 2027 (In Progress)

IN progress

B. Tech, Computer Science & Engineering (Software Engineering) — Ajeenkya DY Patil University, Pune Aug 2022 – Jun 2026

Specialization: Artificial Intelligence, Machine Learning, IoT and Robotics (Minor) | CGPA: 9.24/10.00 | First Class with Distinction

- Key modules: Data Structures & Algorithms, Database Management Systems, Machine Learning, Deep Learning, Operating Systems, Computer Networks, Robotics Programming.
- Maintained First Class with Distinction across all semesters.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Hackathon Leadership

2023

- Captain across multiple national-level hackathons (Smart India Hackathon 2023, Ashna AI Hackathon 2025); led cross-functional technical teams under time pressure.

Robotics Club — Ajeenkya DY Patil University

2022 – Present

- Contributed to autonomous drone and Arduino-based robotic systems projects.

Google Developer Student Club (GDSC)

2023 – 2026

- Organized AI, ML and IoT workshops and mentored junior students.